

2 Muscles, bones and joints enable movement of the skeleton.

(a) (i) Which structure attaches one bone to another bone at a flexible joint?

(1)

- ☐ **A** actin
- ☐ **B** ligament
- ☐ **C** synapse
- ☐ **D** tendon

(ii) How many of the statements about joints are correct?

(1)

- antagonistic pairs of muscles move the bones at a joint
- muscles acting on a joint are myogenic
- tendons are more elastic than ligaments

- ☐ **A** 0
- ☐ **B** 1
- ☐ **C** 2
- ☐ **D** 3

(b) Usain Bolt's world record of 9.58 seconds for the 100m men's sprint is unbeaten.



(Source: © Ian MacNicol / Getty Images)

(i) What is the role of myoglobin in muscle fibres?

(1)

- ☐ **A** it is an enzyme that reacts with myosin
- ☐ **B** it is part of the sarcolemma involved in the release of calcium ions
- ☐ **C** to act as an immediate source of energy
- ☐ **D** to provide oxygen for muscles

(ii) Explain why the muscles of a sprinter have a high percentage of fast twitch fibres.

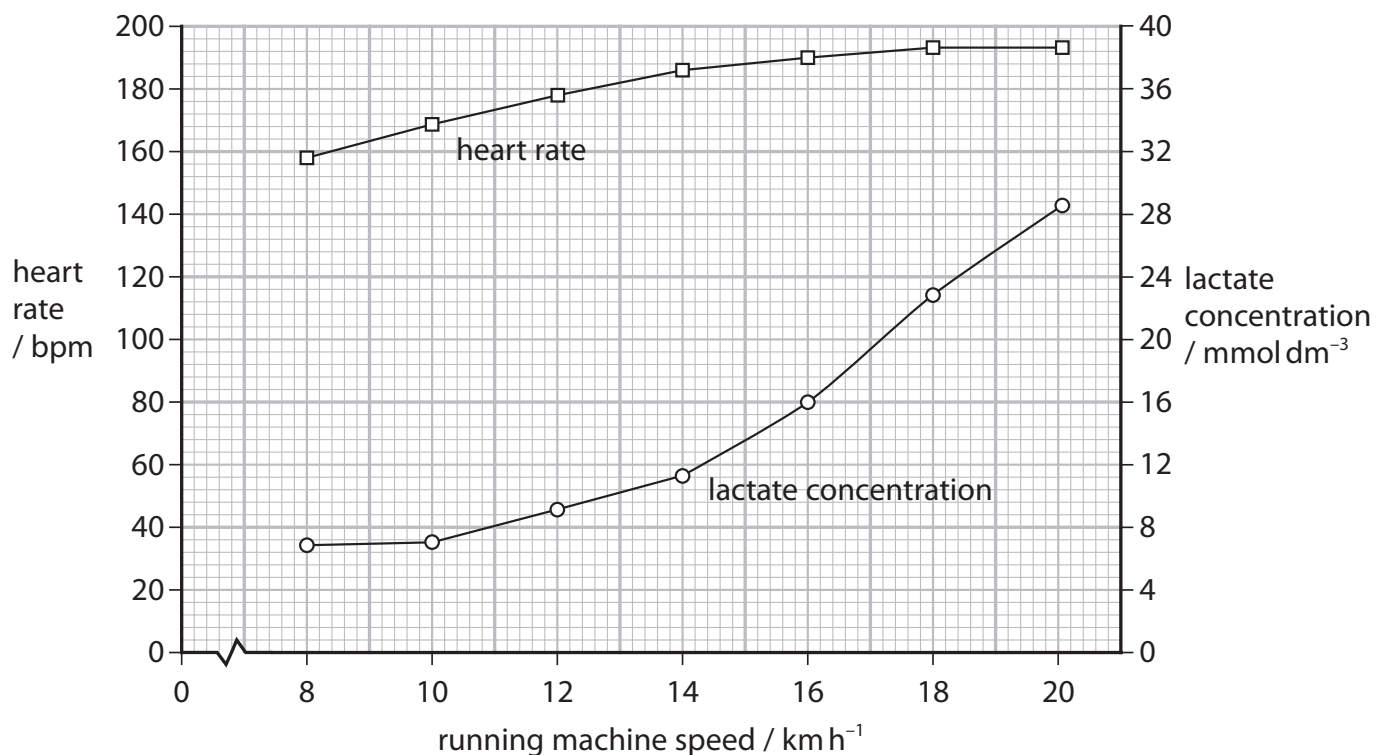
(3)

- (c) In an investigation, an athlete ran for 10 minutes on a running machine moving at constant speed.

The investigation was repeated at seven running speeds.

At each speed, the heart rate and lactate concentration in the blood of the athlete were measured.

The results are shown in the graph.



Comment on the results of this investigation.

(3)

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(Total for Question 2 = 9 marks)